

ETHAN QIU

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EDUCATION

University of Toronto

Sep 2024 - Expected 2028

B.Sc. Computer Science Specialist

- Software Design, Data Structures and Algorithms, Computer Organization, Linear Algebra, Multivariable Calculus

EXPERIENCE

Software Engineer Intern

Apr 2026 - Present

Shopify

- Shipped Shopify's new **variant publishing** and **bulk-edit** flows, resolving **10+** merchant-reported bugs and rolling out to **140,000+** stores.
- Built the **Managed Markets** publishing experience in **React** and **GraphQL**, surfacing sellability status, restriction reasons, and AI-powered explanations across **190+** countries.
- Architected the UI, APIs, and **Sidekick AI** integration for the core prototype of Shopify's **agentic-commerce** Product Details redesign, used by **millions of merchants** to manage their products.

Software Engineer (Co-op)

May 2025 - Aug 2025

General Dynamics Mission Systems-Canada

- Spearheaded a modular **Python automation framework**, reducing regression runtime by **50%**.
- Delivered internal **Java** and **Python** tooling adopted by **3 engineering teams**, eliminating **8 manual test steps**.
- Drove improvements to test coverage and reliability across **15 production PRs**.

Engineering Director - Industry Collaborations

May 2025 - Apr 2026

UTMIST (University of Toronto Machine Intelligence Student Team)

- Led development across **4 industry projects**, coordinating **20+ developers** with weekly sprint cycles.
- Developed an agentic credit card recommendation system using **LangChain** and **vector databases** for Flybits.
- Co-developed **PyTorch** and **XGBoost** models powering pricing decisions for Amicare's home care platform.

Project Lead

May 2025 - Apr 2026

Blueprint, University of Toronto

- Owned delivery of the Museum of Digital Entertainment's digital inventory platform (**50,000+ artifacts**), managing **10 developers** and shipping first release in **6 weeks**.
- Designed scalable **REST APIs**, role-based access control, and optimized **PostgreSQL** schemas.
- Established **CI/CD pipelines** and deployment workflows, ensuring operational reliability for partner teams.

PROJECTS

Heimer Academy - League of Legends Data Analytics | *Python, FastAPI, React, AWS, PostgreSQL* [GitHub]

- Built a scalable cloud-native analytics platform on **AWS**, computing metrics across **2,000+** games.
- Designed distributed ML pipelines using **Amazon SageMaker** and **Bedrock**, executing **100,000+** model calls.
- **1st Place Overall - Rift Rewind AWS Hackathon.**

Identity Matrix - Persistent Multiplayer AI Simulation | *TypeScript, Node.js, FastAPI, Supabase* [GitHub]

- Architected a persistent multiplayer system with human-AI avatar handoff, preserving state across distributed sessions (**1st Place Overall - UofTHacks 2026**).
- Decoupled real-time physics (6-10Hz) from AI inference (1Hz) via event-driven architecture to prevent blocking.
- Implemented probabilistic agent controller combining LLM reasoning with softmax action selection over memory and proximity.

Orbit - Real-Time Social Intelligence System | *React, OpenCV, Whisper, LangChain* [GitHub]

- Created a distributed system with real-time vision, transcripts, and LLM reasoning for contextual awareness.
- Engineered sub-second latency pipelines with persistent memory, indexing **200+ identities** in real time.
- **Best Use of Groq and Windsurf - Hack the North 2025.**

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, Ruby, C/C++, SQL, Swift

Frameworks: React, Ruby on Rails, FastAPI, Django, Node.js, Express, Next.js, GraphQL

Cloud & Tools: AWS (S3, EC2, Lambda, SageMaker, Bedrock), Docker, Xcode, Git, GitHub Actions, PostgreSQL

Libraries: PyTorch, LangChain, OpenCV, pandas, NumPy